Andrew Li

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Education

Columbia University New York, NY
Master of Science in Computer Science *Expected Dec 2026*
- **Selected Coursework:** Distributed Systems, Computer Vision II, Scaling LLMs, High Performance ML

The University of Michigan Ann Arbor, MI
Bachelor of Science Engineering in Computer Science, Minor in Statistics *Sept 2020 – Dec 2023*
- *3.8/4.0 GPA (Summa Cum Laude)*
- **Selected Coursework:** Machine Learning, Operating Systems, Computer Security, Bayesian Data Analysis

Experience

Graduate TA - Scaling LLMs: Systems, Optimization, and Emerging Paradigms Sept 2026 – Dec 2026
Columbia University *New York, NY*
- Mentored 40 graduate students through complex systems challenges, providing hands-on support for debugging GPU kernels, modifying serving systems (vLLM / SGLang), and implementing memory-efficient attention mechanisms like KV cache eviction algorithms.

ML Engineer Intern May 2026 – Aug 2026
Dandy *New York, NY*
- Engineered high-throughput Ray Data streaming pipelines for multi-modal dental datasets, eliminating I/O bottlenecks and GPU starvation across distributed compute nodes.
- Designed and deployed MLflow on GKE, migrating the team away from a legacy Vertex AI and TensorBoard setup to centralize experiment tracking, encourage performance engineering, and improve model reproducibility.
- Diagnosed distributed VRAM bottlenecks in Ray Train on Flyte via CUDA profiling; implemented mixed-precision training and activation checkpointing to scale input capacity 12.5x (8K to 100K inputs).
- Distilled 3 diverse teachers into a single student via soft-voting logits, achieving an 86.5% win rate over baseline while avoiding the 3x inference FLOP penalty.

AI Engineer Feb 2024 – Aug 2025
Epic Systems *Madison, WI*
- Architected internal LLM and agentic workflows (RAG, multi-step chain-of-thought, function calling), enabling schema-constrained output pipelines for applications such as chart summary and hospital course prediction.
- Co-developed an enterprise model evaluation suite to benchmark model statistical performance and fairness across different cohorts, ensuring that AI in healthcare works for patients across all demographics.

FedScale Research Assistant Jun 2022 – Jun 2023
University of Michigan: SymbioticLab *Ann Arbor, MI*
- Researched and implemented secure aggregation, differential privacy, and privacy budgeting algorithms with respect to federated learning.

Projects

FHIRFly: A HIPAA-Compliant Clinical Note Parser | *Python, Pytorch, Transformers* Nov 2025 – Nov 2025
- Built a fast, reliable, and HIPAA-compliant system to extract structured data from unstructured clinical notes.
- Fine-tuned small language models on synthetic data to extract deterministic JSON schemas from noisy unstructured records, while enforcing strict PII redaction and compliance rules.
- Awarded Honorable Mention at AWS-AGI House Hackathon among 30 teams.

Technical Skills

Selected Languages: Python, C/C++, Java, R, MySQL
Selected Tools: vLLM, Ray Data/Train/Serve, CUDA Profiling, Pydantic, Amazon Web Services / GCP, Docker, Kubernetes (GKE), IaC (Pulumi), Temporal, Flyte, MLflow, Weights & Biases
Selected Libraries: NumPy, pandas, scikit-learn, matplotlib, Pytorch, RLZoo, Optuna

Activities and Awards

MIT Battlecode Finalist (placed top 4 seeding, top 7 final), MIT, 2024